Plant nutrition

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Photosynthesis [1]

Which statement describes photosynthesis?

A the process by which plants break down carbohydrates to release energy for growth

B the process by which plants absorb carbohydrates from the soil through their roots

C the process by which plants synthesise carbohydrates from raw materials using energy from light

D the process by which plants take in oxygen and give out carbon dioxide using energy from light

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2 · Part 1 Core Photosynthesis [1]

Chlorophyll is a green pigment.

Which row is correct for chlorophyll?

	where chlorophyll is found	role of chlorophyll
A	chloroplasts	transfers energy from light into energy in chemicals
B	nucleus	transfers energy from light into energy in chemicals
C	chloroplasts	absorbs carbon dioxide from the air spaces in the leaf
D	mitochondria	releases energy from glucose

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3 · Part 1 Core Photosynthesis [1]

Some of the glucose made in photosynthesis is changed into other carbohydrates.

Which row shows the carbohydrate transported in the phloem and the carbohydrate used to build cell walls?

	transported in the phloem	**used to build cell walls**
A	glucose	starch
B	starch	cellulose
C	sucrose	starch
D	sucrose	cellulose

4 · Part 1 Core Photosynthesis [1]

Young bean plants are grown in a solution that contains all the mineral ions they need except magnesium ions.

After three weeks the plants are smaller than plants given all the mineral ions.

Which statement explains why?

A They cannot make amino acids, so less protein is made for growth.

B They make less chlorophyll, so less energy from light is transferred for photosynthesis.

C They cannot make cellulose, because magnesium ions form part of plant cell walls.

D They cannot absorb water, because magnesium ions are needed for osmosis.

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5 · Part 1 Core Photosynthesis [1]

Some leaves have green areas and white areas. The white areas contain no chlorophyll.

One of these plants is kept in the dark for 48 hours. A strip of black paper is then fixed across one of its leaves, covering part of a green area and part of a white area. The plant is placed in bright light for 6 hours. The leaf is then removed and tested for starch with iodine solution.

The table describes four regions of the leaf.

region	colour of leaf in this region	covered by black paper
P	green	no
Q	white	no
R	green	yes
S	white	yes

Which regions turn blue-black?

A P only

B P and Q only

C P and R only

D P, Q and R only

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6 · Part 1 Core Photosynthesis [1]

Hydrogencarbonate indicator solution changes colour as the carbon dioxide concentration changes.

colour of indicator	carbon dioxide concentration
yellow	higher than in air
orange-red	the same as in air
purple	lower than in air

Three test-tubes are set up. Each contains the same volume of orange-red hydrogencarbonate indicator solution and is sealed with a bung.

- Tube 1 contains a piece of pondweed and is placed in bright light.
- Tube 2 contains a piece of pondweed and is wrapped in aluminium foil.
- Tube 3 contains no pondweed and is placed in bright light.

Which row shows the colour of the indicator in each tube after 6 hours?

	tube 1	tube 2	tube 3
A	yellow	purple	orange-red
B	purple	purple	orange-red
C	purple	yellow	orange-red
D	purple	yellow	yellow

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7 · Part 1

Extended | Photosynthesis | [1]

Use the balanced chemical equation for photosynthesis to answer this question.

How many molecules of carbon dioxide are used, and how many molecules of oxygen are released, when a plant makes two molecules of glucose?

	molecules of carbon dioxide used	molecules of oxygen released
A	6	6
B	12	6
C	6	12
D	12	12

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8 · Part 1 Extended Photosynthesis [1]

The rate of photosynthesis of a plant was measured at different light intensities. This was done at two carbon dioxide concentrations. The temperature was kept at 20 °C throughout.

light intensity / arbitrary units	rate of photosynthesis at 0.04% carbon dioxide / arbitrary units	rate of photosynthesis at 0.12% carbon dioxide / arbitrary units
10	8	8
20	16	16
30	21	24
40	22	31
50	22	36

Which factor is limiting the rate of photosynthesis at 0.04% carbon dioxide at a light intensity of 10 arbitrary units, and at a light intensity of 50 arbitrary units?

	at 10 arbitrary units	at 50 arbitrary units
A	carbon dioxide concentration	light intensity
B	light intensity	carbon dioxide concentration
C	light intensity	light intensity
D	carbon dioxide concentration	carbon dioxide concentration

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9 · Part 1 Core Leaf structure [1]

Most leaves are thin.

How does being thin help a leaf to photosynthesise?

A It allows more water to evaporate from the leaf to keep it cool.

B It means each mesophyll cell can contain more chloroplasts.

C It gives a short distance for carbon dioxide to diffuse to the mesophyll cells.

D It reduces the surface area of the leaf so that less water is lost.

10 · Part 1 Core Leaf structure [1]

Which row correctly links a feature of a leaf to how it helps photosynthesis?

	feature of the leaf	how it helps photosynthesis
A	palisade mesophyll cells are just below the upper epidermis	these cells receive the most light
B	spongy mesophyll has large air spaces	the air spaces store water for photosynthesis
C	upper epidermis cells contain many chloroplasts	they absorb light before it reaches the mesophyll
D	xylem is found in the vascular bundles	xylem carries sucrose into the leaf

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11 · Part 1 Core Photosynthesis [6]

Sunflower seedlings are growing in pots on a sunny windowsill.

(a) Describe what is meant by photosynthesis. [2]

(b) State the word equation for photosynthesis. [2]

(c) Name the part of a plant cell where chlorophyll is found. [1]

(d) State the role of chlorophyll in photosynthesis. [1]

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12 · Part 1 Core Photosynthesis [6]

An apple tree makes glucose in its leaves during the summer. The glucose is used in several different ways.

(a) Some of the glucose is changed into sucrose. State why the tree changes glucose into sucrose. [1]

(b) In spring the flowers of the apple tree produce nectar. Explain how producing nectar helps the apple tree to reproduce. [2]

(c) The apple tree loses all its leaves in autumn. In spring, new leaves start to grow before the tree can photosynthesise again. Suggest how the tree obtains the energy it needs for this early growth. [2]

(d) Name the carbohydrate that the growing leaves use to build new cell walls. [1]

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13 · Part 1 Core Photosynthesis [7]

Wheat seedlings were grown in three different nutrient solutions for four weeks. At the start, the mean mass of a seedling in each group was 0.5 g. All other conditions were kept the same.

solution	mineral ions in the solution	mean mass of a seedling after four weeks / g	colour of leaves after four weeks
1	all the mineral ions the plant needs	4.8	dark green
2	all except nitrate ions	1.2	pale green
3	all except magnesium ions	2.0	yellow

(a) Calculate the percentage increase in the mean mass of a seedling grown in solution 1.
Show your working. [2]

(b) Explain why the seedlings in solution 3 have yellow leaves and a lower mean mass than the seedlings in solution 1. [3]

(c) Explain why a lack of nitrate ions reduces the growth of the seedlings in solution 2. [2]

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14 · Part 1 Core Photosynthesis [7]

In an investigation into whether plants need carbon dioxide for photosynthesis, a student used two potted geranium plants of the same size.

- Both plants were kept in a dark cupboard for 48 hours.
- Each plant was then enclosed in a sealed, clear plastic bag.
- The bag around plant X also contained a dish of soda lime. Soda lime absorbs carbon dioxide.
- The bag around plant Y contained an identical dish with no soda lime.
- Both plants were placed in bright light at 25 °C for 6 hours.
- One leaf was then removed from each plant. Each leaf was placed in boiling water, then in hot ethanol, rinsed, and then tested with iodine solution.

(a) Explain why both plants were kept in a dark cupboard for 48 hours before the investigation started. [2]

(b) Suggest why the leaves were placed in hot ethanol before iodine solution was added. [1]

(c) Predict the colour of each leaf after iodine solution was added. Explain your prediction for plant X. [3]

(d) Explain why plant Y was included in the investigation. [1]

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15 · Part 1 Core Photosynthesis [7]

Some pondweed was placed in a beaker of water. A small amount of sodium hydrogencarbonate was added to the water to supply carbon dioxide. A lamp was placed at different distances from the pondweed. At each distance the bubbles of gas given off by the pondweed were counted for one minute. This was repeated three times at each distance.

distance of lamp from pondweed / cm	number of bubbles per minute, trial 1	number of bubbles per minute, trial 2	number of bubbles per minute, trial 3	mean number of bubbles per minute
10	46	50	48	48
20	33	31	35	33
30	19	24	23	
40	12	11	13	12
50	7	6	8	7

(a) Calculate the mean number of bubbles per minute when the lamp was 30 cm from the pondweed. [1]

(b) Describe the effect of the distance of the lamp on the number of bubbles per minute. Use data from the table in your answer. [2]

(c) Explain why the number of bubbles per minute changes as the lamp is moved further away. [2]

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(d) Name the gas in the bubbles. [1]

(e) The lamp also releases heat. Suggest how the student could stop the heat from the lamp affecting the results. [1]

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16 · Part 1 Extended Photosynthesis [8]

The volume of oxygen produced by an aquatic plant in 10 minutes was measured at different temperatures. The light intensity and the carbon dioxide concentration were both kept high throughout.

temperature / °C	volume of oxygen produced in 10 minutes / cm^3
10	0.6
20	1.4
30	2.6
35	3.0
40	2.2
45	0.8

(a) Describe the effect of temperature on the volume of oxygen produced. [2]

(b) Calculate the percentage decrease in the volume of oxygen produced when the temperature increased from 35 °C to 45 °C. Give your answer to the nearest whole number. [2]

(c) Explain why the volume of oxygen produced decreased at temperatures above 35 °C. [2]

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(d) Identify the factor limiting the rate of photosynthesis at 20 °C. Explain your answer. [2]

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17 · Part 1 Core Photosynthesis [6]

Hydrogencarbonate indicator solution shows the carbon dioxide concentration of the solution.

colour of indicator	carbon dioxide concentration
yellow	higher than in air
orange-red	the same as in air
purple	lower than in air

Four sealed tubes were set up, each with the same volume of orange-red hydrogencarbonate indicator solution. Tubes 1, 2 and 3 each contained an equal-sized piece of the same aquatic plant. All four tubes were placed the same distance from a bright lamp for 8 hours.

tube	aquatic plant present	covering around the tube	colour of indicator after 8 hours
1	yes	none	purple
2	yes	several layers of thin cloth	orange-red
3	yes	aluminium foil	yellow
4	no	none	orange-red

(a) Explain the result in tube 3. [2]

(b) Explain the result in tube 1. [2]

(c) Suggest why the colour of the indicator in tube 2 did not change. [1]

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(d) Explain why tube 4 was included. [1]

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18 · Part 1 Extended Photosynthesis [7]

Scientists measured the rate of photosynthesis of wheat leaves at different carbon dioxide concentrations. They did this at a low light intensity and at a high light intensity. The temperature was kept at 25 °C.

carbon dioxide concentration / %	rate of photosynthesis at low light intensity / arbitrary units	rate of photosynthesis at high light intensity / arbitrary units
0.01	6	6
0.02	12	12
0.03	15	18
0.04	16	24
0.06	16	33
0.08	16	38
0.10	16	38

(a) State the balanced chemical equation for photosynthesis. [2]

(b) Identify the factor limiting the rate of photosynthesis between 0.01% and 0.02% carbon dioxide at both light intensities. Use the data to explain your answer. [2]

(c) At low light intensity the rate stays at 16 arbitrary units from 0.04% to 0.10% carbon dioxide. Explain why. [2]

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(d) At high light intensity the rate does not increase between 0.08% and 0.10% carbon dioxide. Suggest one environmental factor, other than carbon dioxide concentration, that could be limiting the rate at 0.10% carbon dioxide. [1]

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19 · Part 1 Extended Photosynthesis [7]

Lettuces are grown in a glasshouse in a country with cold winters. The grower measured the rate of photosynthesis of the lettuces under four sets of conditions.

conditions	temperature / °C	light intensity	carbon dioxide concentration / %	rate of photosynthesis / arbitrary units
1	8	high	0.04	20
2	18	high	0.04	45
3	18	high	0.10	70
4	18	low	0.10	25

(a) Use conditions 1 and 2 to identify the factor limiting the rate of photosynthesis in conditions 1. Explain your answer. [2]

(b) On sunny winter days the grower burns a paraffin heater inside the glasshouse. Burning paraffin releases heat and carbon dioxide. Use the data to explain how burning the paraffin heater increases the growth of the lettuces. [3]

(c) On dull, cloudy days the conditions in the glasshouse are like conditions 4. Suggest what the grower could do to increase the rate of photosynthesis further on these days. Explain your answer. [2]

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20 · Part 1 Core Leaf structure [7]

The leaves of a dicotyledonous plant are adapted for photosynthesis.

(a) Most leaves have a large surface area. Explain how this helps photosynthesis. [1]

(b) The table describes a feature of four structures in the leaf.

structure	feature
upper epidermis	cells are transparent
palisade mesophyll	cells contain many chloroplasts
spongy mesophyll	many air spaces between the cells
stomata	pores in the epidermis

Explain how each feature helps the leaf to photosynthesise.

(i) upper epidermis [1]

(ii) palisade mesophyll [1]

(iii) spongy mesophyll [1]

(iv) stomata [1]

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(c) The vascular bundles in a leaf contain xylem and phloem. Explain how xylem and phloem each help the leaf in photosynthesis. [2]

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